

**OXFORD**

INTERNATIONAL  
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# INTERNATIONAL GCSE PHYSICS

## 9203/1

Paper 1

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Mark scheme

November 2022

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Version: 1.0 Final



2 2 B Y 9 2 0 3 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from [oxfordaqaexams.org.uk](https://www.oxfordaqaexams.org.uk)

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## Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

### Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

### Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

## Information to Examiners

### 1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

### 2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

### 3. Marking points

#### 3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as \* in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

| Student | Response | Marks awarded |
|---------|----------|---------------|
| 1       | green, 5 | 0             |
| 2       | red*, 5  | 1             |
| 3       | red*, 8  | 0             |

Example 2: Name two planets in the solar system.

[2 marks]

| Student | Response                 | Marks awarded |
|---------|--------------------------|---------------|
| 1       | Neptune, Mars, Moon      | 1             |
| 2       | Neptune, Sun, Mars, Moon | 0             |

### 3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

### 3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

### 3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

### 3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

### 3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

### 3.7 Brackets

(.....) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

**3.8 Allow**

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

**3.9 Ignore**

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

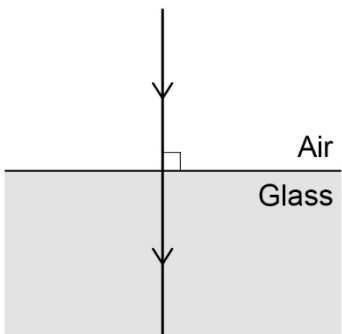
**3.10 Do not accept**

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

**Question 1**

| Question | Answers | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|---------|-------------------|------|-------------------|
| 01.1     | B       |                   | 1    | AO1<br>3.3.5 b    |

| Question | Answers                 | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------------------|-------------------|------|-------------------|
| 01.2     | $n = \frac{0.70}{0.50}$ |                   | 1    | AO2<br>3.3.5e     |
|          | 1.4                     |                   | 1    |                   |

| Question | Answers                                                                             | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------------------------------------------------------------------------------|-------------------|------|-------------------|
| 01.3     |  |                   | 1    | AO1<br>3.3.5 a    |

| Question | Answers                                | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|----------------------------------------|-------------------|------|-------------------|
| 01.4     | to detect light at the back of the eye |                   | 1    | AO1<br>3.3.6 j    |

| Question | Answers              | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|----------------------|-------------------|------|-------------------|
| 01.5     | eyeball is too short |                   | 1    | AO1<br>3.3.6 j    |

| Question | Answers | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|---------|-------------------|------|-------------------|
| 01.6     | convex  |                   | 1    | AO1<br>3.3.6 j    |

| Question | Answers | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|---------|-------------------|------|-------------------|
| 01.7     | light   |                   | 1    | AO1<br>3.3.6 k    |

| Question | Answers                    | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|----------------------------|-------------------|------|-------------------|
| 01.8     | to cauterise blood vessels |                   | 1    | AO1<br>3.3.6 k    |

|                  |  |   |
|------------------|--|---|
| Total Question 1 |  | 9 |
|------------------|--|---|



**Question 2**

| Question    | Answers          | Extra information                                        | Mark | AO/<br>Spec. Ref. |
|-------------|------------------|----------------------------------------------------------|------|-------------------|
| <b>02.1</b> | use a water bath | allow Bunsen burner<br>allow any sensible heating method | 1    | AO4<br>3.4.1 e    |
|             | wear goggles     | allow any sensible safety precaution                     | 1    |                   |

| Question    | Answers     | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|-------------|-------------------|------|-------------------|
| <b>02.2</b> | stopclock   | allow stopwatch   | 1    | AO4<br>3.4.1 e    |
|             | thermometer |                   | 1    |                   |

| Question    | Answers | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|---------|-------------------|------|-------------------|
| <b>02.3</b> | 60 s    |                   | 1    | AO2<br>3.4.1 e    |

| Question    | Answers   | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|-----------|-------------------|------|-------------------|
| <b>02.4</b> | 24.5 (°C) |                   | 1    | AO2<br>3.4.1 e    |

| Question    | Answers                | Extra information                   | Mark | AO/<br>Spec. Ref. |
|-------------|------------------------|-------------------------------------|------|-------------------|
| <b>02.5</b> | $29.5 - 24.5 = 5$ (°C) | allow ecf from Question <b>02.4</b> | 1    | AO2<br>3.4.1 e    |

| Question    | Answers                     | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|-----------------------------|-------------------|------|-------------------|
| <b>02.6</b> | $E = 0.080 \times 110\,000$ |                   | 1    | AO2<br>3.4.1 d    |
|             | 8 800 (J)                   |                   | 1    |                   |

| Question    | Answers                                               | Extra information            | Mark | AO/<br>Spec. Ref. |
|-------------|-------------------------------------------------------|------------------------------|------|-------------------|
| <b>02.7</b> | rate of energy transfer decreases                     | allow cools down more slowly | 1    | AO3<br>3.4.2 d    |
|             | as the temperature difference with the room decreases |                              | 1    |                   |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 2</b> |  | <b>11</b> |
|-------------------------|--|-----------|

**Question 3**

| Question | Answers                                         | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------------------------------------------|-------------------|------|-------------------|
| 03.1     | metal <b>B</b> expands more than metal <b>A</b> |                   | 1    | AO1<br>3.4.2 e    |

| Question | Answers                                               | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------------------------------------------------|-------------------|------|-------------------|
| 03.2     | iron <b>and</b> zinc                                  |                   | 1    | AO3<br>3.4.2 e    |
|          | because they have the largest difference in expansion |                   | 1    |                   |

| Question | Answers                                               | Extra information                | Mark | AO/<br>Spec. Ref. |
|----------|-------------------------------------------------------|----------------------------------|------|-------------------|
| 03.3     | percentage increase<br>$= \frac{2.2}{700} \times 100$ |                                  | 1    | AO2<br>3.4.2 e    |
|          | 0.31 (%)                                              | allow answer that rounds to 0.31 | 1    |                   |

| Question | Answers                                                             | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|---------------------------------------------------------------------|-------------------|------|-------------------|
| 03.4     | when temperature rises, bimetallic strip bends to touch the contact |                   | 1    | AO1<br>3.4.2 e    |
|          | this completes the circuit to turn on the fire alarm                |                   | 1    |                   |

| Question | Answers                      | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|------------------------------|-------------------|------|-------------------|
| 03.5     | particles vibrate more       |                   | 1    | AO3<br>3.4.2 e    |
|          | particles move further apart |                   | 1    |                   |

|                         |  |          |
|-------------------------|--|----------|
| <b>Total Question 3</b> |  | <b>9</b> |
|-------------------------|--|----------|

**Question 4**

| Question    | Answers                                                | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|--------------------------------------------------------|-------------------|------|-------------------|
| <b>04.1</b> | number of complete waves passing a point in one second |                   | 1    | AO1<br>3.3.1 g    |

| Question    | Answers           | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|-------------------|-------------------|------|-------------------|
| <b>04.2</b> | 24 000 000 000 Hz |                   | 1    | AO2<br>3.3.1 g    |

| Question    | Answers                                                | Extra information                   | Mark | AO/<br>Spec. Ref. |
|-------------|--------------------------------------------------------|-------------------------------------|------|-------------------|
| <b>04.3</b> | 300 000 000 =<br>24 000 000 000 × wavelength           | allow ecf from Question <b>04.2</b> | 1    | AO2<br>3.3.1 h    |
|             | wavelength = $\frac{300\,000\,000}{24\,000\,000\,000}$ |                                     | 1    |                   |
|             | 0.0125 (m)                                             | allow 0.013 (m)                     | 1    |                   |

| Question    | Answers                                                                                                                                                                                                                                                                                             | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------|-------------------|
| <b>04.4</b> | any <b>two</b> from: <ul style="list-style-type: none"> <li>• travel at same speed (through a vacuum)</li> <li>• transverse waves</li> <li>• can be refracted</li> <li>• can be reflected</li> <li>• can be diffracted</li> <li>• transfer energy</li> <li>• can travel through a vacuum</li> </ul> |                   | 2    | AO1<br>3.3.2 b    |

| Question    | Answers                                                                                       | Extra information                                           | Mark | AO/<br>Spec. Ref. |
|-------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|-------------------|
| <b>04.5</b> | light waves have a shorter wavelength<br><br><b>or</b><br>light waves have a higher frequency | allow different wavelength<br><br>allow different frequency | 1    | AO1<br>3.3.2 b    |

| Question    | Answers                                                                                                                                                                                                                  | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------|-------------------|
| <b>04.6</b> | any <b>two</b> from:<br><br><ul style="list-style-type: none"> <li>• condition of tyres</li> <li>• condition of brakes</li> <li>• icy / wet roads</li> <li>• condition of road surface</li> <li>• mass of car</li> </ul> |                   | 2    | AO1<br>3.1.5 e    |

| Question    | Answers                                                                                                                                                                                                                                                                                                                        | Extra information | Mark                | AO/<br>Spec. Ref. |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------|-------------------|
| <b>04.7</b> | the stopping distance is the sum of the thinking distance and the braking distance<br><br>(because) the thinking distance is decreased as the distance travelled during the driver's reaction time is decreased<br><br>(and) the braking distance is decreased as the distance travelled during the braking force is decreased |                   | 1<br><br>1<br><br>1 | AO1<br>3.1.5 b    |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 4</b> |  | <b>13</b> |
|-------------------------|--|-----------|

**Question 5**

| Question    | Answers                                                                                        | Extra information                            | Mark | AO/<br>Spec. Ref. |
|-------------|------------------------------------------------------------------------------------------------|----------------------------------------------|------|-------------------|
| <b>05.1</b> | it can be switched on and off<br><br>or<br><br>the strength of the electromagnet can be varied | allow can be stronger than permanent magnets | 1    | AO3<br>3.5.2 g    |

| Question    | Answers                                                                          | Extra information                                      | Mark       | AO/<br>Spec. Ref. |
|-------------|----------------------------------------------------------------------------------|--------------------------------------------------------|------------|-------------------|
| <b>05.2</b> | increase the current in the coil<br><br>increase the number of turns in the coil | allow use a battery with a higher potential difference | 1<br><br>1 | AO1<br>3.5.2 g    |

| Question    | Answers                                                                           | Extra information | Mark | AO/<br>Spec. Ref.                        |
|-------------|-----------------------------------------------------------------------------------|-------------------|------|------------------------------------------|
| <b>05.3</b> | (when the switch is closed)<br>there is a current in the electromagnet circuit    |                   | 1    | 2 × AO1<br>2 × AO2<br>2 × AO3<br>3.5.2 c |
|             | producing a magnetic field around the electromagnet                               |                   | 1    |                                          |
|             | the iron bar becomes an induced magnet                                            |                   | 1    |                                          |
|             | iron bar is attracted to the electromagnet                                        |                   | 1    |                                          |
|             | pulling the brake pad onto the wheel so friction acts between brake pad and wheel |                   | 1    |                                          |
|             | work done against friction so kinetic energy is reduced                           |                   | 1    |                                          |

|                         |  |          |
|-------------------------|--|----------|
| <b>Total Question 5</b> |  | <b>9</b> |
|-------------------------|--|----------|

**Question 6**

| Question    | Answers                                   | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|-------------------------------------------|-------------------|------|-------------------|
| <b>06.1</b> | two hydrogen nuclei                       |                   | 1    | AO1<br>3.7.4 a b  |
|             | join / fuse to form a helium nucleus      |                   | 1    |                   |
|             | some of the mass is converted into energy |                   | 1    |                   |

| Question    | Answers                                                                                                    | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|------------------------------------------------------------------------------------------------------------|-------------------|------|-------------------|
| <b>06.2</b> | force of repulsion between two positive nuclei needs to be overcome (so they can get close enough to fuse) |                   | 1    | AO1<br>3.7.4 c    |

| Question    | Answers                                    | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|--------------------------------------------|-------------------|------|-------------------|
| <b>06.3</b> | the star will expand to become a red giant |                   | 1    | AO1<br>3.8.1 h    |
|             | the outer layers of gas will be shed       |                   | 1    |                   |
|             | exposing the core as a white dwarf         |                   | 1    |                   |
|             | this will cool to become a black dwarf     |                   | 1    |                   |

| Question    | Answers                                                                              | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|--------------------------------------------------------------------------------------|-------------------|------|-------------------|
| <b>06.4</b> | the range of electromagnetic radiation emitted by an object at a certain temperature |                   | 1    | AO1<br>3.3.2 f    |

| Question                | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Mark      | AO/<br>Spec. Ref. |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------|
| <b>06.5</b>             | <b>Level 2:</b> Scientifically relevant features are identified; the way(s) in which they are similar / different is made clear and (where appropriate) the magnitude of the similarity / difference is noted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3–4       | AO3<br>3.3.2 f    |
|                         | <b>Level 1:</b> Relevant features are identified and differences noted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1–2       |                   |
|                         | <b>No relevant content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0         |                   |
|                         | <b>Indicative content</b> <ul style="list-style-type: none"> <li>• blue star has maximum intensity at 400 nm</li> <li>• white star has maximum intensity at 620 nm</li> <li>• red star has max intensity at 790 nm</li> <li>• the greater the surface temperature of the star the shorter the wavelength at its peak intensity</li> <li>• the blue star would be the brightest</li> </ul> <p>for all stars:</p> <ul style="list-style-type: none"> <li>• as wavelength increases, intensity increases then decreases</li> </ul> <p>for all wavelengths:</p> <ul style="list-style-type: none"> <li>• blue star has greatest intensity</li> <li>• red star has least intensity</li> </ul> |           |                   |
| <b>Total Question 6</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>13</b> |                   |



**Question 7**

| Question    | Answers                                                                                                                                                                                                                         | Extra information                                           | Mark | AO/<br>Spec. Ref. |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|-------------------|
| <b>07.1</b> | any <b>one</b> from: <ul style="list-style-type: none"> <li>• uses a renewable energy resource</li> <li>• (operation of the fountain) does not contribute to global warming</li> <li>• reduces risk of electrocution</li> </ul> | (operation of the fountain) does not produce carbon dioxide | 1    | AO1<br>3.2.3 d    |

| Question    | Answers                                         | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|-------------------------------------------------|-------------------|------|-------------------|
| <b>07.2</b> | $0.30 = \frac{12\,000}{\text{total energy in}}$ |                   | 1    | AO2<br>3.2.2 f    |
|             | $\text{total energy in} = \frac{12\,000}{0.30}$ |                   | 1    |                   |
|             | 40 000 (J)                                      |                   | 1    |                   |

| Question    | Answers                | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|------------------------|-------------------|------|-------------------|
| <b>07.3</b> | $6.0 = \frac{E}{3000}$ |                   | 1    | AO2<br>3.5.1 f    |
|             | $E = 6.0 \times 3000$  |                   | 1    |                   |
|             | 18 000 (J)             |                   | 1    |                   |

| Question    | Answers                                                | Extra information                      | Mark | AO/<br>Spec. Ref. |
|-------------|--------------------------------------------------------|----------------------------------------|------|-------------------|
| <b>07.4</b> | $E_k = 0.5 \times 2.0 \times 10^{-4} \times 3.5^2$     |                                        | 1    | AO2<br>3.2.1 d e  |
|             | $E_k = 0.001225 \text{ J}$                             |                                        | 1    |                   |
|             | $E_k = E_p$                                            |                                        |      |                   |
|             | $0.001225 = 2.0 \times 10^{-4} \times 9.8 \times h$    | allow $h = \frac{v^2}{2g}$             | 1    |                   |
|             | $h = \frac{0.001225}{(2.0 \times 10^{-4} \times 9.8)}$ | allow $h = \frac{3.5^2}{2 \times 9.8}$ | 1    |                   |
|             | 0.625 (m)                                              | allow 0.63 (m)                         | 1    |                   |

| Question    | Answers                          | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|----------------------------------|-------------------|------|-------------------|
| <b>07.5</b> | work done against air resistance |                   | 1    | AO1<br>3.2.1 b    |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 7</b> |  | <b>13</b> |
|-------------------------|--|-----------|

**Question 8**

| Question    | Answers                                                    | Extra information                                                            | Mark | AO/<br>Spec. Ref.                                 |
|-------------|------------------------------------------------------------|------------------------------------------------------------------------------|------|---------------------------------------------------|
| <b>08.1</b> | section <b>A</b> to <b>B</b>                               | allow speed for velocity throughout                                          |      | 2 x AO1<br>4 x AO2<br>3.1.3 b c d<br>g<br>3.1.6 a |
|             | the resultant force is forwards                            | allow thrust is greater than drag                                            | 1    |                                                   |
|             | so the car accelerates                                     | allow velocity increases                                                     | 1    |                                                   |
|             | section <b>B</b> to <b>C</b>                               |                                                                              |      |                                                   |
|             | resultant force is zero                                    | allow thrust and drag have the same magnitude but act in opposite directions | 1    |                                                   |
|             | so the car travels at a constant velocity                  | allow terminal velocity for constant velocity                                | 1    |                                                   |
|             | section <b>C</b> to <b>D</b>                               |                                                                              |      |                                                   |
|             | the resultant force is in the opposite direction to motion | allow drag is greater than thrust                                            | 1    |                                                   |
|             | so the car decelerates                                     | allow velocity decreases for decelerates                                     | 1    |                                                   |

| Question    | Answers                      | Extra information                                           | Mark | AO/<br>Spec. Ref.         |
|-------------|------------------------------|-------------------------------------------------------------|------|---------------------------|
| <b>08.2</b> | $a = \frac{45 - 0}{1.4 - 0}$ | allow use of any point on line <b>AB</b>                    | 1    | AO1<br>2 x AO2<br>3.1.3 e |
|             | 32                           | allow a correct answer to any number of significant figures | 1    |                           |
|             | m/s <sup>2</sup>             |                                                             | 1    |                           |

| Question | Answers                                             | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-----------------------------------------------------|-------------------|------|-------------------|
| 08.3     | <b>distance travelled during acceleration</b>       |                   |      | AO2<br>3.1.3 f    |
|          | $0.5 \times 137.5 \times 4.3$                       |                   | 1    |                   |
|          | 295.625 (m)                                         |                   | 1    |                   |
|          | <b>distance travelled during constant speed</b>     |                   |      |                   |
|          | $137.5 \times 2.2 = 302.5$ (m)                      |                   | 1    |                   |
|          | total distance = $295.625 + 302.5$<br>= 598.125 (m) |                   | 1    |                   |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 8</b> |  | <b>13</b> |
|-------------------------|--|-----------|